

Education

- **Institute of Science and Technology Austria (ISTA)** Sep. 2021 - expected Sep. 2026
Ph.D. candidate in Computer Science, supervised by Dan Alistarh
Topic: Efficient training of neural networks
- **Sharif University of Technology, Iran** Sep. 2016 - Jul. 2021
B.Sc. in Computer Engineering

Research Experience

- **Ph.D. Candidate at ISTA** Sep. 2021 - present
 - Published *HALO*, *QuEST*, and *Quartet*, enabling accurate pretraining/finetuning of LLMs directly in quantized INT8/FP6/MXFP4 formats with significant training and inference speedups.
 - Introduced *Robust Adaptation (RoSA)*, a parameter-efficient fine-tuning algorithm inspired by robust principal component analysis, improving upon low-rank adaptation (*LoRA*).
 - Proposed *SparseProp*, the first algorithm to leverage unstructured sparsity for faster neural network training. Complemented it by efficient CPU implementations achieving significant training speedups (3.6x on 95% sparse ResNet18).
- **Student Researcher at Google Research** Oct. 2024 - Jun. 2025
 - Worked on efficient data selection for LLMs under Vincent Cohen-Addad and Vahab Mirrokni.
 - Published *Influence Distillation*, an efficient data selection method for targeted instruction tuning at scale. It allows up to 20× data compression with little to no accuracy loss, while offering up to 3.5× faster selection compared to state-of-the-art at the time.
- **Research Intern at EPFL, Switzerland** Jul. 2020 - Dec. 2020
 - Implemented cross-modal contrastive learning and test-time training techniques to identify and mitigate distribution shifts in vehicle trajectory prediction. Supervised by Alexandre Alahi.
- **Research Assistant at Sharif University of Technology, Iran** Jul. 2019 - Mar. 2020
 - Under the supervision of Mahdieh Soleymani, introduced *3D-SegReg* for weakly-supervised 3D medical image segmentation, resulting in a notable 7% increase in F1-score on the CHAOS challenge compared to the leading method at the time, with reduced supervision requirements.

Honors and Awards

- **Iran's National University Entrance Exam** Sep. 2016
Ranked among top 0.1% in Mathematics and Physics.
- **Iran's National Mathematical Olympiad** Sep. 2015
Won a silver medal. Awarded National Elites Foundation Fellowship.

Other Projects

- **Panza Email Assistant, ISTA** Mar. 2024 - Jun. 2024
Published *Panza* (in a team of nine), a personal email assistant that can be privately trained on a single 16-24GB GPU. [github](#)
- **Industry Internship at Quera Co., Iran** Mar. 2021 - Jun. 2021
Designed an online Git course (in a team of three), purchased by 2000+ users in Iran.
- **Trade Management Plugin Development** Jan. 2021 - Apr. 2021
Created *Nutricula* (in a team of four), a MetaTrader plugin offering intuitive and flexible trade management solutions. *Nutricula* received acclaim from numerous trading experts in Iran and currently has 1000+ premium users.

Languages/Technical Skills: Python (torch, JAX), C++, AVX/AVX2, Git, Java, MQL, L^AT_EX

Hobbies: Chess (1800 FIDE Elo rating), Volleyball

Publications

- **M. Nikdan**, V. Cohen-Addad, D. Alistarh, and V. Mirrokni, “Efficient Data Selection at Scale via Influence Distillation,” Preprint 2025.
- R. L. Castro*, A. Panferov*, S. Tabesh, O. Sieberling, J. Chen, **M. Nikdan**, S. Ashkboos, and D. Alistarh, “Quartet: Native FP4 Training Can Be Optimal for Large Language Models” Preprint 2025, Spotlight at ICML ES-FoMo-III Workshop.
- A. Panferov, J. Chen, S. Tabesh, R. L. Castro, **M. Nikdan**, and D. Alistarh, “QuEST: Stable Training of LLMs with 1-Bit Weights and Activations,” ICML 2025.
- S. Ashkboos*, **M. Nikdan***, S. Tabesh*, R. L. Castro, T. Hoefler, and D. Alistarh, “HALO: Hadamard-Assisted Lossless Optimization for Efficient Low-Precision LLM Training and Fine-Tuning,” Preprint 2025.
- A. Nicolicioiu*, E. Iofinova*, A. Jovanovic*, E. Kurtic*, **M. Nikdan***, A. Panferov, I. Markov, N. Shavit, and D. Alistarh, “Panza: Design and analysis of a fully-local personalized text writing assistant.” Preprint 2024.
- **M. Nikdan***, S. Tabesh*, E. Crnčević, and D. Alistarh, “RoSA: Accurate Parameter-Efficient Fine-Tuning via Robust Adaptation.” ICML 2024.
- **M. Nikdan***, T. Pegolotti*, E. Iofinova, E. Kurtic, and D. Alistarh, “SparseProp: Efficient Sparse Backpropagation for Faster Training of Neural Networks.” Oral at ICML 2023, and Spotlight in Sparsity in Neural Networks (SNN) Workshop 2023.
- A. Bitarafan, **M. Nikdan**, and M. Soleymani Baghshah, “3D Image Segmentation with Sparse Annotation by Self-Training and Internal Registration,” in IEEE Journal of Biomedical and Health Informatics 25.7: 2665-2672, 2020.